



GLOBAL ECONOMICS
GRADE: 11TH
TERM 3 – LEARNING EVIDENCE 1
HYPOTHESIS TEST DESIGN (C1 – C3)
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**THIRD
TERM
2025 – 2026**

Learning objective: Design hypothesis tests from real contexts by identifying the correct population parameter, setting and justifying a benchmark claim and significance level, and determining the test tail direction from the contextual claim.

Evaluated criteria:

C1: Describes the concept of a statistical hypothesis.
 C2: Identifies the null and alternative hypotheses in a statistical test.
 C3: Determines whether a test should be one-tailed or two-tailed.

Criteria assessment

Each assessment criterion is evaluated through the worked examples and activity questions. A criterion is considered passed when it is correctly applied in 4 of the 6 assigned tasks.

For a set of tasks to count as valid and complete, it must include

- exactly 3 proportion/probability settings and exactly 3 mean settings, so that Criterion C1 can be assessed properly
- at least 1 left-tailed setup (lower than), at least 1 right-tailed setup (higher than), and at least 1 two-tailed setup (different from), so that Criteria C2 and C3 can be assessed properly

Flashcards with the meaning of the different symbols are allowed. However, all flashcards will be checked. If a flashcard contains a solved or developed problem, the exam will be nullified. In addition, no exams will be accepted after the established submission time. Any late submission, even if it is only 5 seconds late, will not be received.

Student Problems

Use the following shared instructions for all six problems:

(C1) **Describe the context**; state whether it is a population mean or population proportion setting; **match the parameters** with their corresponding symbols; **explain how you would collect sample information**; and invent a plausible sample result.

(C2) **State the benchmark claim and justify it narratively**; choose a significance level; then write H_0 and H_a with correct notation, keeping the null as “parameter equals benchmark.”

(C3) **Classify your test** as left-tailed, right-tailed, or two-tailed, **draw the corresponding graph**, and **justify the classification** from the wording and intention of your claim.

1. Problem 1 — School context

Create a hypothesis test about a measurable feature of your school.

2. Problem 2 — Home context

Create a hypothesis test about a measurable feature of life at home.

3. Problem 3 — Neighborhood context

Create a hypothesis test about a measurable feature of your neighborhood.

4. Problem 4 — Your Grade 11 group context

Create a hypothesis test about a measurable feature of your own Grade 11 group.

5. Problem 5 — Personal context

Create a hypothesis test about a measurable feature related to yourself.

6. Problem 6 — Bogotá context

Create a hypothesis test about a measurable feature of Bogotá that can be investigated at your level.